

Sets

Definition: A set is a collection of objects (known as elements)

Sets are written using $\{$ and $\}$

e.g., set of Natural numbers = $\{1, 2, 3, \dots\}$

set of the names James, James, and Jacob = $\{\text{James}, \text{Jacob}\}$

Duplicates or orders don't matter

An element of which is a set is not the same as the elements contained within the set

Definition: The **cardinality** of a set is the number of elements contained in a finite set (or infinite)
Defined using $|A|$ where A is a set

e.g., let $A = \{1, 3, 9, 7, 3\}$
 $|A| = 5$

$\in$ means 'in' or 'is an element of'

Sets can be constructed from other defined sets

They are written as $\{\text{set to build on} : \text{properties of elements in set}\}$

e.g., Set of integers between -1 and $3 = \{x \in \mathbb{Z} : -1 < x < 3\}$

Common Sets

$\mathbb{N}$ = natural numbers = $\{1, 2, 3, \dots\}$

$\mathbb{Z}$ = integers = $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

$\mathbb{Q}$ = rationals = $\{\frac{a}{b} : a, b \in \mathbb{Z} \text{ and } b \neq 0\}$

$\mathbb{R}$ = real numbers * From Cantor's Diagonalisation argument, we find that you cannot actually define the real numbers using a set

$\emptyset$ = the empty set

$\mathbb{C}$ = complex numbers = $\{x+iy : x, y \in \mathbb{R}\}$

Linking Sets

Definition: let B be a set. A is a subset of B if all of the elements in A , are also in B

This is written as $A \subseteq B$

e.g., $A = \{1, 2, 3, 4, 5\}$

$\therefore A \subseteq \mathbb{N}$

Definition: Sets A and B are considered equal if $A \subseteq B$ and $B \subseteq A$